

DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No- 13	Team No- 15	Project Title- Online Food Delivery Platform
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1. Abstract

The Online Food Delivery Platform is a full-stack web application that digitizes the complete food ordering experience by connecting three types of users on a single system: restaurants, customers, and delivery partners. Traditional phone-based ordering offers no live tracking, no centralized payments, and no organized delivery coordination. This project addresses that gap by allowing restaurant owners to manage their menus, customers to browse restaurants, place orders and pay online, and delivery partners to accept and fulfill deliveries, all coordinated through a real-time backend.

The proposed solution uses Socket.io to push live order status and delivery-partner location updates, and the Google Maps API to visualize real-time tracking on a map. Payments are processed securely through Razorpay, and all data is persisted in a MySQL database. An admin panel oversees restaurant onboarding, commission tracking, and platform-wide order management. As a key differentiator from existing platforms such as Swiggy and Zomato, this project introduces Group Ordering with Live Split Bill, allowing multiple users to add items to a shared cart in real time and automatically split the payment among participants via Razorpay. Together, these components recreate a working ecosystem similar to commercial food delivery apps, while giving hands-on experience with real-time systems, geolocation, and payment gateway integration.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520090201	MOSANGI VIJAYA LAKSHMI	
2520090211	POTLABATHINI SHASHANTH	
2520090102	D PAWAN KUMAR	

3. Frameworks / Platforms / Software Used

(List the frameworks, libraries, platforms, or tools you plan to use.)

Frontend: React.js (Customer, Restaurant & Delivery Partner apps)

Backend: Node.js, Express.js

Database: MySQL

Real-time Communication: Socket.io (live order status & delivery tracking)

Location Services: Google Maps API

Payments: Razorpay

Key Differentiator: Group Ordering with Live Split Bill – a real-time shared cart (via Socket.io) that lets multiple users on one order add items simultaneously, with the final bill auto-split and collected per person through Razorpay. This feature is not offered by existing platforms like Swiggy or Zomato.

Guide Name and Signature (remarks if any) _____